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The role of the customer order decoupling point in production and supply chain management

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ABSTRACT

In order to compete successfully, operations in any type of firm need to be strategically aligned to the market requirements. This concerns both manufacturing and supply chain operations. The customer order decoupling point (CODP) is getting increasing attention as an important input to the design of manufacturing operations as well as supply chains. This paper investigates the impact of the position and role of the CODP on issues of concern for production and supply chain management. The focus is on the design and strategic planning aspects of the supply chain, and the design of manufacturing planning and control systems. The paper proposes a dual design approach for production and supply chain planning systems; one type of system for operations upstream the CODP and another type of system for downstream operations in order to fully support the characteristics and objectives of each respective part of the supply chain.

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1. Introduction

In order to compete successfully, operations in any type of firm need to be strategically aligned to the market requirements. The motivation for this research is that companies are showing increasing interest in incorporating the customer order decoupling point (CODP) as an important input to the strategic design of manufacturing operations as well as supply chains. The CODP is the point in the material flow where the product is tied to a specific customer order; the basic choices being make-to-stock, assemble-to-order, make-to-order, and engineer-to-order. As a rule, the CODP coincides with the most important stock point, from where the customer order process starts. From a supply chain perspective, there is typically one dominant CODP along the entire supply chain flow. From a company perspective, the CODP can be positioned inside their manufacturing operations or it can be positioned at the suppliers (first tier or even further upstream in the supply chain), at the interface with the supplier (raw material inventory), at the border towards the customers (at some finished goods inventory), or even further downstream in the supply chain.

This paper investigates the impact of the position and role of the customer order decoupling point on production and supply chain management. Issues that will be dealt with include the design and planning of manufacturing and supply chain operations. The purpose of the paper is to explore and develop a framework for an integrated perspective on production and supply chain management based on the CODP. The development of the framework in this paper is based on three types of research sources: a review of the related literature, a case study, and the results from an empirical survey of 128 Swedish manufacturing firms.

First, a review is provided on the related literature on the customer order decoupling point and issues related to production and supply chain management that can be interpreted from a CODP perspective. Secondly, the case study and the survey results are presented. Then, the framework is presented and motivated.

2. Related literature

2.1. The customer order decoupling point

The CODP is traditionally defined as the point in the value chain for a product, where the product is linked to a specific customer order. Sometimes the CODP is called the order penetration point [1,2]. Different manufacturing situations such as make-to-stock (MTS), assemble-to-order (ATO), make-to-order (MTO) and engineer-to-order (ETO) all relate to different positions of the CODP. The different situations are related to the ability of manufacturing operations to accommodate product customization or a wide product range; cf. Fig. 1. Thereby, the CODP divides the material flow that is forecast-driven (upstream of the CODP) from the flow that is customer order-driven (downstream the CODP).

The CODP is the point where product specifications get frozen in most cases, and more important, it is the last point at which inventory is held [1]. Thus, the inventory at the CODP is a strategic

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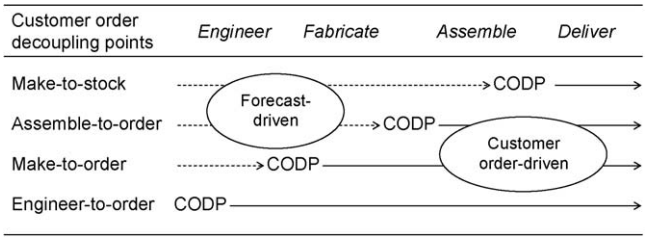


Fig. 1. Different customer order decoupling points (based on Sharman, 1984 [1]).

stock point since delivery promises are based on the stock availability at the CODP and the lead times and capacity availability for the customer order-driven operations downstream the CODP [2]. The literature explicitly discussing the CODP is growing, dealing with both manufacturing and supply chain operations. In [3], the CODP is considered a major factor in designing an integrated push-pull manufacturing strategy. More recent contributions are found in [4–13]. There is a strong consensus among the literature on CODP in that the operations upstream are significantly different than those downstream, based on the fact that the upstream material flow is forecast-driven, whereas real customer orders dominate downstream. Publications primarily from the Cardiff group [7,8,10] distinguished between lean and agile supply chains. This literature acknowledges the CODP as the divider between lean and agile operations in manufacturing or supply chains. A lean supply chain should be applied upstream the CODP, while an agile supply chain would be more suitable for downstream operations. This is the core idea of the "leagility" approach.

2.2. Managerial differentiation with respect to manufacturing planning and control

The models for linking market characteristics to the design of manufacturing planning and control (MPC) systems can basically be reduced to one, first presented Berry and Hill in [14], and later included in [15,16]. This model is concerned with three hierarchical planning levels: master planning (make-to-order, MTO; assemble-to-order, ATO; make-to-stock, MTS), material planning (time-phased vs. rate-based), and shop floor control (MRP-type vs. JIT-type; i.e. material requirements planning vs. just-in-time). The fourth level in a manufacturing resource planning (MRPII) system structure is sales and operations planning, which was added to the Berry and Hill framework by [17], concerned with the choice of planning strategy (chase vs. level, or potential mixes thereof; cf. [2]).

The attributes linking market and product characteristics to the planning and control system are basically the same at all levels, i.e. product type, product range and individual product volume per period [14,18]. Firms with high-volume standardized products are assumed to utilize a level planning strategy, make-to-stock, rate-based, and pull approaches, whereas firms with many low-volume, customized products are expected to choose a chase planning strategy, make-to-order, time-phased, and push approaches [2,14–17]. In ATO environments these choices are applicable to different sections of the production system or the supply chain, with respect to the position of the CODP; cf. the leagility literature, implying that lean principles are applied upstream the CODP while agile principles are applied downstream. Volumes are typically sufficiently high before the CODP to make level/MTS/rate-based/pull possible, whereas chase/MTO/time-phased/push is typically required after the CODP due to customized features and low volumes per product. The applicability of Kanban and MRP to different manufacturing environments is supported in [19], stating that "overall, pull systems, such as Kanban, may work well when the demand variability is low" and that MRP is a logical choice for "a firm producing a wide range of custom products in small batches using a make-to-order policy". They further suggested that a careful choice of the MPC system to fit the manufacturing environment will help improve the overall performance of the firm. The results of an empirical study of 211 high performing plants in seven countries show that lean principles are significantly associated with high levels of cost performance, while agile principles are better aligned with flexibility performance [20]. This result supports the perception that lean approaches are preferable upstream the CODP and agile approaches downstream.

2.3. Managerial differentiation with respect to supply chain design

Fisher presented a framework for choosing the right supply chain for products in [21]. He made a distinction between physically efficient and market-responsive supply chains. Products are categorised as either functional or innovative and should be matched to the proper supply chain delivery system. Functional products characterized by, e.g. a steady demand pattern and long product life cycles should be managed in a physically efficient supply chain that focuses on cost minimization and high utilization of resources, whereas innovative products with demand volatility and short life cycles should be transformed through a market-responsive supply chain that has extra capacity, capability of market demand information processing, and that is more flexible. Thus, this model suggested that certain product characteristics require a specific type of supply chain design, while other choices lead to mismatches. The model has attracted considerable attention by both practitioners and researchers. Other researchers have proposed models that build upon the model by Fisher, such as in [22–24]. The characteristics of the physically efficient supply chain are applicable to operations upstream the CODP, whereas the characteristics of the market-responsive supply chain are useful for downstream operations. Another perspective on supply chain management with relevance to the CODP is the supply chain operations reference (SCOR) model [25]. The SCOR model differentiates the three basic processes source, make, and deliver with respect to make-to-stock, make-to-order, and engineer-to-order products. Thus, the SCOR model acknowledges that the position of the decoupling point has an impact on the design of operations processes.

3. Case study

The case company is operating in the telecommunication industry, and manufactures microwave radio solutions for applications in telecom networks. The case study is concerned with the major product line. The informants to the case study are two supply chain coordinators, one market coordinator, one manager from operational procurement and production logistics. A detailed account of the case study and methodology is provided in [26]. There are approximately 1000 variants of the main product line offered to the marketplace. With all auxiliary equipment such as antennas and power supply, the number of possible configurations of complete microwave radio systems is over one million. In the upstream stages of the production, the number of variants is much more limited. The overall supply chain focus of the firm is to respond quickly to unpredictable demand. The sales and operations planning function manages the mix of forecasting for upstream operations and suppliers and of customized orders for the final production stages.

The main internal and external value streams are depicted in Fig. 2. Suppliers deliver printed circuits, components and modules. Components are surface mounted on printed circuit boards and modules are finalised into multi-chip modules and supplied to the

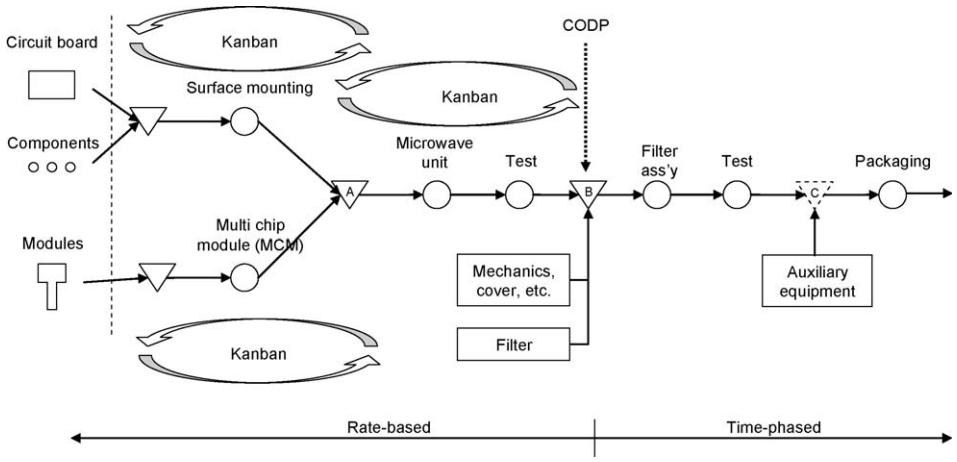


Fig. 2. Production and supply chain structure of the case company (Based on Selldin, 2005 [26]).

intermediate inventory A. The circuit boards and multi-chip modules are assembled to form a microwave unit. The microwave units are tested before being delivered into inventory B. Filters needed to give each unit the specified radio frequency are supplied from outside suppliers to the inventory B. Mechanical parts and covers are also supplied to this inventory. B is the CODP from where non-standard products with highly fluctuating demand are produced based solely on customer orders. The microwave unit is equipped with a filter to give it a unique frequency. The complete microwave unit with the filter is then mounted in a protective cover. The complete assembly is tested in different temperature conditions and supplied to inventory C. The products are then packed with all other equipment such as modem units and antennas and shipped to customers.

A Kanban system is used to control the flow of material from the raw material inventories up to the CODP. The company use three different Kanban loops; one for the surface mounting in order to replenish inventory A, one for the multi-chip module for replenishment at inventory A, and one for the manufacture and test of the microwave unit to replenish inventory B. The section between inventories B and C is normally treated as a make-to-order section. However, when capacity utilization at the filter assembly is low, items with relatively stable and frequent demand can be produced to stock at inventory position C. Still, the standard operating procedures use the CODP at inventory position B as the focal point for differentiating the planning and control approaches. This is due to the fact that the section between B and C must have time-phased planning capabilities. Time-phased approaches can be made to behave like a rate-based system with respect to lot sizing and repetitive schedules and sequences, while the reverse is not possible (i.e. a rate-based system cannot be made to work like a time-phased system). The company has developed a hybrid planning approach that includes Kanban-type functionality with a push system.

The case company makes a clear differentiation between upstream and downstream operations, emphasizing the importance of taking the CODP into strategic consideration when designing and managing production and supply chain operations. A main consideration is the characteristics of the products or items that are managed at each stage of the production and supply chain. The main characteristics concerning how upstream and downstream operations are differentiated at the case company are summarized in Table 1.

4. Survey findings

The aspects on the CODP relative to supply chain issues and planning and control issues have been tested empirically in [18,27]. Here, a summary of the findings with specific relevance to the CODP is presented. The study is based on a mail questionnaire survey, with responses from 128 Swedish manufacturing companies; see [18,27] for a full account of the empirical study.

4.1. Supply chain design issues

The framework by Fisher [21] is tested in [27]. The two types of products (functional and innovative) as well as the two types of supply chains (physically efficient and market-responsive) were verified based on a factor analysis of product related items and supply chain related items, respectively. The factors that most clearly relate to the framework by Fisher were: (i) product life cycle and lead time required for functional products, (ii) average margin of forecast error and average stock-out rate for innovative products, (iii) minimize cost and minimize inventory for physically efficient supply chains, and (iv) excess buffer capacity and excess buffer stocks for market-responsive supply chains. When comparing combinations of product type and supply chain type, some statistical significance was noted. The results showed that the combination of functional products and physically efficient supply chains was significant in two ways. Companies with functional products chose a physically efficient supply chain rather than a market-responsive (significant at the 0.01 level), and companies with a physically efficient supply chain used it for functional products rather than for innovative products (significant at the 0.01 level).

Table 1
Some key differences observed in the case study.

Aspect	Upstream the CODP	Downstream the CODP
Product	Standard components, high volumes, predictable demand	Fully customized, unpredictable demand
Process	Assembly line type	Job shop type
Supply chain	Physically efficient, high utilization, low-cost, supplier selection based on cost and lead time	Market-responsive, buffer capacity
Planning and control	Make-to-stock, Kanban	Make-to-order, customer orders

In summary, the findings indicated that it is possible to distinguish two types of products and two types of supply chains. Relating the findings to the CODP, the physically efficient supply chain fits the characteristics of the operations upstream the CODP and the market-responsive supply chain fits the downstream part. Also, the type of products associated with each part fits the type of product according to the expectations around the CODP.

4.2. Manufacturing planning and control issues

The framework by Berry and Hill [14] is tested in [18], capturing three hierarchical planning levels, with the extension of the sales and operations planning level added by [17]. The empirical study showed that the choices at the sales and operations planning level (chase vs. level) and the master planning level (make-to-stock, assemble-to-order, and make-to-order) were significantly related to the product characteristics, e.g. product demand volume. It was further shown that these choices had a positive significant mediating impact on operational performance, in terms of delivery speed, delivery reliability, volume flexibility and product mix flexibility. Thus, fitting the planning approaches for the two upper MPC levels to the type of product had a positive and significant impact. The planning approaches at the two lower levels (material planning and production activity control) were not significantly related to either product characteristics or performance. It was observed that time-phased material planning as well as MRP-like production activity control approaches were used in both dynamic and static environments. Two explanations were offered: (i) time-phased and MRP-like approaches work well in any manufacturing environment and (ii) firms are conservative in their use of MPC approaches in that they stay with MRP by tradition even though they have the market characteristics that would allow them to use rate-based techniques. However, the reverse was not observed, since rate-based and JIT-type approaches seemed to be more related to static manufacturing environments; rate-based approaches are based on a periodic rate (daily or weekly), traditionally applied to high-volume and process industries [28]. Thus, the empirical results of the study in [18] provided support for the upper two levels (sales and operations planning, and master planning), but not for the lower two in the Berry and Hill model. A possible explanation for the lack of significance for the two lower MPC levels is that material planning and production activity control are related to and concerned with the volumes and repetitiveness of individual items rather than end products (that experience the market requirements).

Relating these findings to the CODP, high-volume items should have a level planning strategy for sales and operations planning and a make-to-stock approach for master planning, which fit the part of the supply chain upstream the CODP. Furthermore, low-volume items should follow a chase planning strategy and a make-to-order approach for master planning, which fit the downstream part of the supply chain. Thus, both the type of product and the MPC approaches fit the structure for differentiating the supply chain at the CODP.

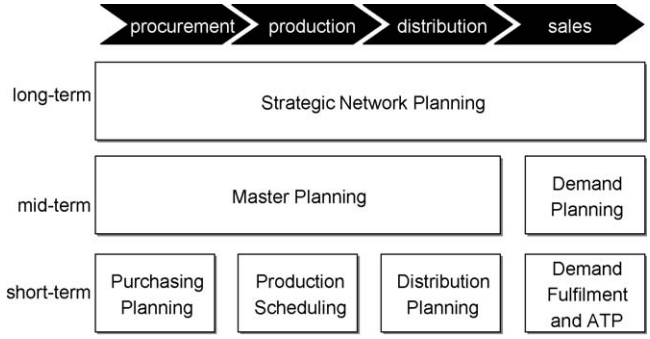


Fig. 3. The supply chain planning matrix (based on Stadtler and Kilger, 2000 [29]).

5. Framework

5.1. A framework for production and supply chain management around the decoupling point

Stadtler and Kilger introduced a matrix approach for the structure of advanced planning systems in [29]; cf. Fig. 3. It includes four stages in a supply chain, i.e. procurement, production, distribution, and sales. The structure holds for an individual firm as well as an entire supply chain.

The modules in the supply chain planning (SCP) matrix correspond largely to the previous discussion on factors that are relevant from a CODP perspective. Table 2 provides a summary of the key production and supply chain planning characteristics upstream (MTS-oriented) and downstream (MTO-oriented) operations from an MPC system perspective. The design and planning at all levels of the supply chain is different for upstream and downstream operations, implying that the SCP system needs to be designed differently for these two parts. The system has to take the CODP into account in order to fit the system to the task. Therefore, a dual approach to the design of SCP systems is necessary, including strategic network planning and planning and control approaches at other levels. All these planning areas need to take into account the type of approach that is appropriate for the particular type of operations—upstream or downstream the CODP. Fig. 4 illustrates the idea of such a dual design approach, where the two systems are based on the characteristics in Table 2. This approach is supported by the case study and the results of the empirical survey, and is well aligned with the extended Berry and Hill framework.

An area of specific interest is the CODP and the activities around it. The main objective at the CODP is to maintain the right mix of products in the right quantities. The demand management

Table 2
Some important manufacturing and supply chain characteristics of upstream and downstream operations.

Attributes	MTS and pre-CODP operations	MTO and post-CODP operations
Strategic network planning	Physical efficiency, based on lean principles	Market-responsiveness, based on agile principles
Master planning	Make-to-stock, replenishment of CODP inventory position	Make-to-order, delivery planning of customer orders with respect to capacity
Demand planning	Forecasting of items in the CODP inventory	Customer order management, product customization
Demand fulfilment	ATP, based on stock availability	ATP, based on lead time agreement, material and capacity availability
Distribution planning	The CODP is the most important inventory of the entire supply chain[a]	No inventory, all activities are driven by actual customer demand
Production scheduling	Rate-based approaches, e.g. Kanban	Time-phased approaches, e.g. MRP, with respect to capacity
Purchasing planning	Rate-based approaches, e.g. Kanban	Time-phased approaches, e.g. MRP

[a] There may be other stock points in the distribution system in a make-to-stock situation further downstream and closer to the customer, e.g. distribution centres, warehouses and retail stores.

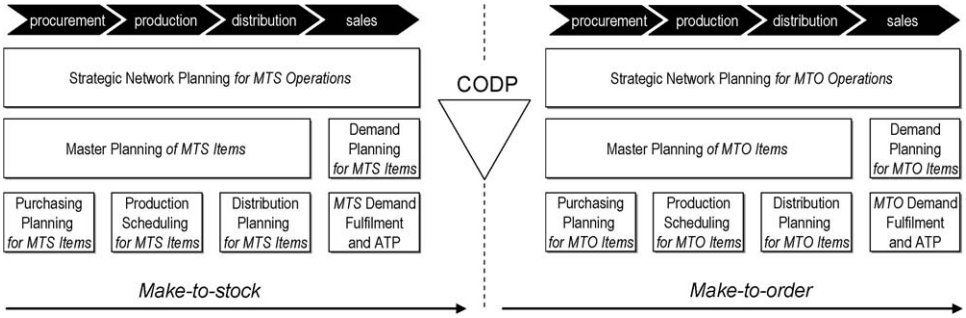


Fig. 4. Manufacturing and supply chain planning matrix differentiation around the CODP; cf. planning characteristics in Table 2.

activities of the upstream part aim to fulfil this requirement. The master planning of the downstream part basically assumes that the correct quantities and items are available in the decoupling inventory at the CODP position. An alternative is to use a two-level master plan that includes both the material availability at the CODP level and the capacity availability of the finishing operations, implying an active planning of the CODP. The purchasing planning is time-phased since the requirements from the finishing stages are time-phased. This means that the withdrawal of items at the CODP inventory is time-phased with respect to specific product configurations, but that the upstream replenishment process can be rate-based since the overall demand is sufficiently high to motivate that the item is kept in stock.

5.2. Deployment of market requirements

One consequence of the role of the CODP is for the deployment of market requirements. Operations that are pure MTS, pure MTO, and pure ETO can transfer the perception of market requirements to all internal operations, since the CODP is positioned outside the manufacturing activities. In ETO, the CODP may well be at a supplier site. Thus, all activities for the focal manufacturing firm will have the same type of attributes and there will be no conflict of priorities (such as between make-to-order and make-to-stock products). If the operations have an ATO character, i.e. an internal CODP, the market requirements only apply to the operations downstream the CODP. This part is MTO by nature, but delivery promises are based on the presumption that the right material is available at the CODP. Flexibility is focussed and lead time reduction is required to continuously improve delivery speed and dependability. Upstream the CODP the operations should be characterised by the attributes for MTS operations, focussing on productivity and cost reduction to continuously improve the low-cost production capabilities. Thus, another mechanism is required than just deploying market requirements to all operations. If market requirements are deployed to all operations – even those upstream the CODP – the upstream part will not be able to support the competitive priorities of the end product, i.e. ensuring that the inventory at the CODP contains the right quantity of the right items at all times and at a competitive price. As a consequence, poor delivery performance can be expected. Therefore, a mechanism has to be designed to ensure that all products having an internal CODP will have the best support from all operations along the supply chain. Therefore, the CODP must be given strategic attention and it must be acknowledged that the stock position at the CODP has a set of competitive priorities that differ from the market requirements. The CODP acts as a make-to-stock controlled stock point, implying that upstream operations should use the product requirements at the CODP position, which consequently become the "market" requirements of the CODP. The only exception is when the CODP is positioned at the ultimate consumer, since all operations will be of make-to-stock character.

An advantage of using the CODP as a point for differentiation is that the choice of MTS versus MTO is typically a natural and clear-cut one in practice, and that the differences and consequences are easily understood by management. For example, the specialty chemical firm Rohm and Haas separated the products into MTS and MTO categories based on demand volume and variability [30].

6. Conclusion

This paper investigated the role of the CODP for production management and supply chain management issues. The focus of this paper has been the key role of the CODP in differentiating production or supply chain operations, by making a distinct decoupling of the operations upstream and downstream the CODP. Building on the models by Fisher [21], linking the choice of supply chain to the product type, and Berry and Hill [14], linking the choice of MPC system to the product type, and other related literature, a dual design approach for manufacturing and supply chain planning systems is proposed.

The major insights from this study are twofold. Firstly, the production and supply chain management system has to be fitted to the task. This means that the systems design at the strategic, tactical as well as the operational level has to consider the position of the CODP. Secondly, the market requirements need to be transformed at the CODP. The CODP acts as a decoupling mechanism separating the customer order-driven activities from those that are forecast-driven. At the CODP the character of the operations change, such that another design strategy is required for the upstream part, wherefore the end-customer market requirements cannot be used as an input for the strategic planning of these operations. The "market" at the CODP has its own characteristics (such as competitive priorities) that the operations upstream the CODP have to acknowledge and address to successfully provide the right products in the right quantities at the right cost level at the CODP inventory position. With this in place, the downstream operations have the prerequisites for providing the marketplace with customized products in a timely and effective manner.

These results are generally applicable to manufacturing and supply chain operations. For firms where there is only one type of decoupling situation, i.e. only MTO or only MTS, the firm can apply a single approach for designing the manufacturing and supply chain planning system. However, most firms tend to have a mix of MTO and MTS products, wherefore both types of systems need to be developed for different parts of the firm. Also, in ATO situations the two types of systems need to be applied to different parts of the supply chain for a single product line. The fact that the entire supply chain is not aligned towards one goal (i.e. the competitive priorities of the ultimate consumer) is not a dilemma. The important issue is to fit the planning and control approach to the task of each respective material flow—both upstream and downstream the CODP.

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